

Magnetic Particle Inspection (MPI) Procedure

90%

Pass

Honeywell

✓ AC Yoke Method

✓ 1. Objective and Application

✓ Objective:

The aim of this inspection is to identify surface and near-surface defects such as cracks, seams, and laps in ferromagnetic materials.

✓ Inspection Method:

Portable AC electromagnetic yoke technique is used.

Reference Standards:

Applicable codes include ASME Section V Article 7, ASTM E709, and DS-1 requirements.

✓ 2. Tools and Consumables

✓ Equipment:

Portable AC electromagnetic yoke unit

✓ Performance Verification:

Yoke must be capable of lifting at least 4.5 kg (10 lb) at the widest pole spacing used during inspection

✓ Magnetic Media:

Dry magnetic powder

Wet visible or fluorescent magnetic particles

✓ Auxiliary Materials:

White contrast paint (used when black magnetic ink or particles are applied)

Cleaning materials such as solvents and wire brushes

✓ Lighting Requirement:

Minimum white light intensity of 1000 lux (100 foot-candles) is recommended

✓ 3. Inspection Procedure

✓ 3.1 Surface Cleaning and Preparation

Ensure removal of oil, grease, rust, paint scale, and dust.

The inspection surface must be clean and completely dry before testing.

✓ 3.2 Application of Contrast Paint (If Required)

Apply a thin and uniform layer of white contrast coating.

This improves visibility of magnetic particle indications.

✓ 3.3 Magnetization Using Yoke

Position the yoke legs on the component covering the inspection zone.

Switch ON the yoke to create a magnetic field between the poles.

Since the magnetic field flows pole-to-pole, rotate the yoke approximately 90° to detect defects in all directions (longitudinal and transverse).

✓ 3.4 Application of Magnetic Particles

Apply particles while magnetization current is ON (continuous technique).

For wet particles, allow excess carrier fluid to drain so particles remain at defect locations.

✓ 3.5 Examination and Assessment

Inspect the surface immediately for particle build-up (indications).

Compare indications with applicable acceptance criteria as per relevant codes or standards.

✓ 4. After Inspection Activities

✓ Demagnetization:

Normally not necessary for AC yoke testing.

Perform only if required by specification or client requirement.

✓ Cleaning:

Remove contrast coating and residual magnetic particles.

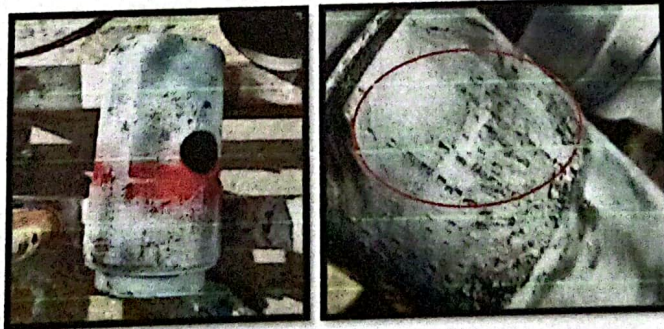
✓ Documentation:

Record inspection results including defect type, size, and exact location.

MAGNETIC PARTICLE INSPECTION REPORT

Customer	✓ DOWELL SCHLUMBERGER SA		Date of Work	✓ 14-05-2026
Rig & Well No.	---		W.O. No.	9493
Location	D & M		Cert. No.	✓ PTIS-MPI-9493-001
Material / Item Type and Description	SHAFT COUPLING			
Material / Item serial No.	FS96RB0773			
Type of Inspection	✓ MAGNETIC PARTICLE INSPECTION		Standard / Specification	✓ ASTM-E 709
Date of Inspection.	14-05-2026		PTIS SOP Reference:	
			Validity of Inspection:	
Surface	✓ Buffing	☐ Grinding	✓ Wire Brush	☐ Machining
Preparation	☐ As Blasted	✓ Removed Paint	☐ Part Dismantled	✓ Part Not Dismantled
MAGNETIZATION UNIT :	AC Yoke		Serial No.	✓ ACY-21
Technique	✓ Yoke	Lifting Capacity	10 Pound	Calibration Certificate No
		Pole Spacing	2 - 6 Inches	Calibration Date:
		Method	Continuous Method	Calibration Due Date:
				✓ PTIS-ACY-21-25-3
				✓ 28-12-2025
				✓ 27-06-2026

Testing Medium	Dry (Iron Powder) ☐	Wet ☒	Fluorescent ☐
Particle Type	Nabakem SM-15 ✓	Colour : Black ✓	Application : Spray
Contrast Medium	Nabakem MP-35 ✓	Colour : White ✓	Drying Time : 5 mins.
Consumable Traceability			
Inspection Consumable	Manufacturer	Batch No.	Expiry Date
White Contrast Paint	Nabakem	240724/1C	03-09-2027
Black Ink	Nabakem	240719/7	03-09-2027



Area of inspection:	FULL BODY MPI INSPECTION ✓
Results :	FOUND TO BE BODY CRACKS AT THE TIME OF INSPECTION.
Recommendation/ Comments:	NOT SATISFACTORY FOR USE: /
INSPECTED BY:	NAND KUMAR NAND KUMAR MOHANTY Inspector Level II METPTIT

